# Surgical outcomes of aortic valve repair for specific aortic valve cusp characteristics; retraction, calcification, and fenestration

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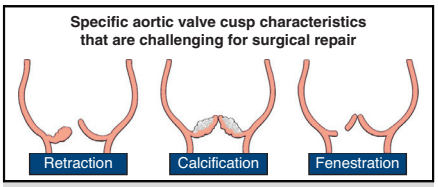
## ABSTRACT

**Objectives:** We investigated the predictive value of aortic valve cusp retraction, calcification, and fenestration for aortic valvuloplasty feasibility.

**Methods:** Multicenter data were collected for 2082 patients who underwent surgical aortic valvuloplasty or aortic valve replacement. The study population had retraction, calcification, or fenestration in at least one aortic valve cusp. Controls had normal or prolapsed cusps.

**Results:** All cusp characteristics demonstrated significantly increased odds ratios [ORs] for switch to valve replacement. This effect was strongest for cusp retraction, followed by calcification and fenestration (OR, 25.14; $P \leq .001$; OR, 13.50, $P \leq .001$; OR, 12.32, $P \leq .001$). Calcification and retraction displayed increased odds for developing grade 4 aortic regurgitation compared with grade 0 or 1 combined on average over time (OR, 6.67; $P \leq .001$; OR, 4.13; $P = .038$). Patients with cusp retraction showed increased risk for reintervention at 1- and 2-year follow-up after aortic valvuloplasty (hazard ratio, 5.66; $P \leq .001$; hazard ratio, 3.22, $P = .007$). Cusp fenestration was the only group showing neither an increased risk of postoperative severe aortic regurgitation ($P = .57$) or early reintervention ($P = .88$) compared with the control group.

**Conclusions:** Aortic valve cusp retraction, calcification, and fenestration were all related to increased rates of switch to valve replacement. Calcification and retraction were associated with recurrence of severe aortic regurgitation. Retraction was related to early reintervention. Fenestration was neither associated with recurrence of severe aortic regurgitation or reintervention. This indicates that surgeons are well able to distinguish aortic valve repair candidates in patients with cusp fenestration. (J Thorac Cardiovasc Surg 2023;166:1627-34)




**Challenging aortic valve cusp phenotypes for surgical aortic valve repair.**

### CENTRAL MESSAGE

Aortic valve cusp calcification and retraction are relative contraindications for aortic valve repair. In case of cusp fenestration, surgeons distinguish the right cases for aortic valve repair.

### PERSPECTIVE

This study reinforces the notion that specific cusp characteristics can influence outcomes of aortic valvuloplasty. Cusp calcification and retraction were associated with recurrence of severe aortic regurgitation, whereas cusp retraction was associated with aortic valve–related reintervention. Better understanding of these characteristics could aid in deciding upon aortic valve repair feasibility.

Aortic regurgitation (AR) refers to diastolic reversed blood flow from the aorta into the left ventricle.[1] Incidence and severity of AR increases with age and is more common in men.[2] Severe AR is treated surgically with aortic valvuloplasty (AVP) or aortic valve replacement (AVR). Although AVR is performed regularly in patients with AR, this procedure has its limitations. Complications regarding valve durability, thromboembolic events, and

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**Abbreviations and Acronyms**

| | |
|---|---|
| AR | = aortic regurgitation |
| AV | = aortic valve |
| AVP | = aortic valvuloplasty |
| AVR | = aortic valve replacement |
| EuroSCORE | = European System for Cardiac Operative Risk Evaluation |
| GEE | = generalized estimating equation |
| HR | = hazard ratio |
| OR | = odds ratio |
| SD | = standard deviation |

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excessive bleeding remain important risk factors for patients with mechanical or biological prostheses in surgical AVR.[3] Valve-conserving procedures such as AVP can reduce these risks. The literature suggests that AVP is superior over AVR, with fewer complications reported, and therefore it is the preferred treatment in patients with AR.[4-7]

The classification of AR typically is based on the assessment of cusp leaflets and aortic root anatomy. This classification system consists of 3 types: type I involves normal aortic valve (AV) cusps with functional aortic annulus dilatation; type II is characterized by cusp prolapse; and type III is defined by diminished tissue quality combined with restricted cusp motion.[8,9] Based on the type of AR and aspects of the aortic root, various surgical approaches are possible.[10] Patients with AR with normal AV cusps or cusp prolapse may be excellent candidates for AVP.[11]

Various cusp characteristics requiring challenging AVP procedures can appear in patients with AR. Well-known variations among these include cusp retraction, calcification, and fenestration (Figure 1). Patients with type III AR, indicating retraction, proved to be less-suitable candidates for AVP. Cusps are considered retracted when the geometric height is ≤16 mm in tricuspid or ≤19 mm in bicuspid aortic valves.[12] Severe AV cusp calcification also appears to be a contraindication for surgical AVP because of its association with disease progression leading to late valve dysfunction.[13-15] In addition, cusp fenestration also has been suggested to be a predictor of valve repair failure.[16]

Pathologic cusp variations might lead to less-favorable clinical outcomes in patients undergoing AVP (Figure 2). However, to date, only few studies have investigated the association between specific cusp characteristics and clinical outcomes after AVP. Therefore, the aim of this study is to assess clinical outcomes after AVP in patients with AR with abnormal AV cusp morphologies, compared with patients with normal AV cusps. Better insight in the clinical predictive value of specific cusp characteristics might aid surgeons in deciding upon AVP feasibility and help prevent complications.

## PATIENTS AND METHODS

### Patient Population

This prospective cohort study included patients with specific AV cusp characteristics selected from the multicenter Aortic Valve Database of the Heart Valve Society.[17] This international registry was established to promote collaboration between hospitals, facilitate knowledge-sharing, and provide sufficient patient numbers for research. Individual informed consent was waived for each patient enrolled in the database at each participating center. Approval was obtained from the Amsterdam University Medical Center Ethics Committee (reference: 15.0236, date: July 22, 2015). Individual informed consent was waived for each participating patient in this study. The study investigated adult patients (>18 years) who underwent surgical AVP or AVR between March 1996 and June 2021. The study included AV cusp phenotypes such as normal AV cusps, prolapse, retraction, calcification, or fenestration, which were assigned based on echocardiographic findings. None of the patients included in the study had a history of previous cardiac surgery. Patients who had cusp vegetation, combined cusp pathology (combinations of cusp fenestration, calcification, and/or retraction within the AV) and those with missing echocardiographic or follow-up data were excluded from the study. As a result of these criteria, follow-up rate was 100%. The selection process resulting in the final enrollment of 2082 participants is illustrated in Figure E1.

Participants were divided into a control and study group based on AV cusp characteristics. Patients with normal cusps or prolapsed cusps were assigned to the control group, whereas patients with cusp retraction, calcification, or fenestration were attributed to the study group. Age- and sex-matching procedures were performed for both groups. The study group was divided into 3 subgroups according to type of AV cusp pathology. This resulted in a total of 4 groups: control, retraction, calcification, and fenestration.

Patients with normal cusp characteristics and cusp prolapse were combined as controls because several studies reported comparable surgical results for AVP in these patients.[18-20] Patients with a combination of different cusp pathologies were excluded from the analysis as this would add another level of complexity to the statistical model.

### Surgical Techniques

Before surgery, included patients were evaluated for AVP feasibility based on clinical characteristics and echocardiographic assessment.[21] Depending on the mechanism of AR, different surgical approaches were used. This could be AV cusp repair (ranging from leaflet plication to running sutures, pericardial patches, decalcification and cusp resection), AV annuloplasty, valve-sparing root replacement, or a combination of these techniques.[22-24] In some patients, reducing the sinotubular junction with a ring was necessary.[25]

### Statistical Analyses

Continuous variables are presented with corresponding means and standard deviations (SDs) or as median with interquartile range when a normal distribution could not be assumed. Binary and categorical variables are reported as frequencies or percentages. To investigate differences in baseline

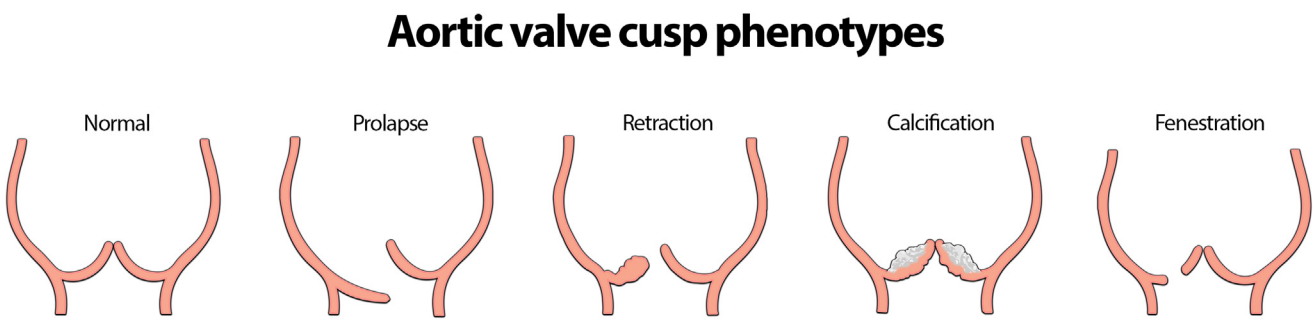


**FIGURE 1.** Aortic valve cusp phenotypes.

characteristics between the 4 groups, univariate general linear models were used for normally distributed data, whereas nonparametric Kruskal–Wallis tests were used for non-normally distributed data. Categorical variables were compared using $\chi^2$ testing.

A follow-up time of 1 and 2 years was used to assess AV-related surgical reintervention after surgical AVP. A time-to-event analysis was performed using the Kaplan–Meier method with Mantel–Cox log rank testing. Patients were censored if no AV-related reintervention occurred or if follow-up time was below the chosen follow-up period. Afterwards, Cox regression analyses were used to investigate differences between the 4 groups (control, retraction, calcification, and fenestration). Possible effect modification was investigated for sex, age, and valve morphology. Results were adjusted for sex, age, valve morphology and European System for Cardiac Operative Risk Evaluation (EuroSCORE) II[26] at baseline. In addition, a Kaplan–Meier and Cox regression time-to-event analysis was performed in the same manner to assess freedom from severe AR to determine durability of AVP.

To assess the impact of cusp characteristics on AR recurrence after AVP, logistic generalized estimating equation (GEE) analyses were performed. For every patient, an echocardiogram at discharge and an echocardiogram at follow-up was provided. AR grade was categorized into 5 ordinal groups, ranging from absent to severe. AR grades 0 (absent) and 1 (mild) were combined as reference group. Three separate logistic GEE analyses were performed (grade 0 + 1 vs 2, grade 0 + 1 vs 3, and grade 0 + 1 vs 4). For every patient, the highest AR grade over time was considered for analysis. Results were adjusted for sex, age, EuroSCORE II, and number of days after initial surgery.

To evaluate the predictive value of cusp pathology in deciding upon AVR despite initial intention to repair, a logistic GEE analysis was performed. A subject-variable at center level was used to correct for possible correlated observations in addition to adjustments for confounders. All statistical analyses were performed using IBM SPSS Statistics, version 28.0, software for Windows (IBM Corp).

## RESULTS

This study included 2082 patients with a mean age of 53.87 years (SD 13.67). Patients with AV cusp retraction were found to be significantly younger, with a mean age of 49.30 years ($P \leq .001$) (Table 1). In total, 84% of the subjects were male, with significant differences observed between control and study groups ($P \leq .001$). Cusp fenestration was less common, whereas cusp retraction was more common in female patients. Significant differences were also observed in valve morphology ($P \leq .001$), presence of rheumatic diseases ($P \leq .001$), and EuroSCORE II ($P \leq .001$). A total of 1881(90.3%) patients underwent AVP. Aside from patients with cusp retraction, the majority of AVPs were performed using a valve-sparing root replacement (N = 929; $P \leq .001$). Valve repair with plicating stiches was reported most frequently among participants ($P \leq .001$), whereas running sutures were least common ($P \leq .001$).

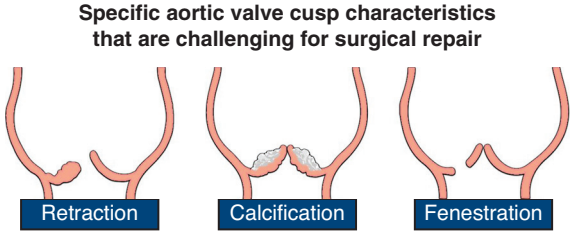


**FIGURE 2.** Aortic valve cusp retraction, calcification and fenestration are challenging phenotypes for surgical aortic valve repair.

### Complications

Complications at 2-year follow-up are summarized in Table 2. Thromboembolic events (strokes, transient ischemic attacks, and peripheral embolisms) were scarce in the control group (N = 11; 0.1%) and were more common in patients with cusp calcification (N = 6; 1.2%) and AV cusp fenestration (N = 4; 1.0%). Although AV-related reinterventions appeared frequently in patients with cusp retraction, rarely any other complication was reported in this subgroup (major bleeding = 1; pacemaker implantation = 1).

### Switch to AVR

Out of 2082 patients undergoing AV surgery, 1867 had a preoperative intention for AVP. Surgeons converted to AVR in 68 cases (3.6%) (Table 3). Conversions predominately occurred in the study groups (control = 6, fenestration = 24, calcification = 25, retraction = 13). Results show increased odds of switching to AVR after perioperative assessment of AV leaflets for every cusp characteristic compared with the control group. This effect was strongest for patients with cusp retraction (odds ratio [OR], 25.14; 4.84-130.57; $P \leq .001$), followed by cusp calcification (OR, 13.50; 3.30-55.15; $P \leq .001$) and cusp fenestration (OR, 5.12.32; 4.47-33.90; $P \leq .001$). In 59 patients, surgeons converted to AVR during the first

**TABLE 1. Preoperative and operative characteristics of patients undergoing surgical aortic valvuloplasty or aortic valve replacement**

| Patient characteristics | Control (N = 1041) | Study (N = 1041) | | | *P* value |
|---|---|---|---|---|---|
| | Normal (N = 301) and prolapse (N = 740) | Fenestration (N = 412) | Calcification (N = 485) | Retraction (N = 144) | |
| Age, y (SD) | 53.87 (13.67) | 53.75 (13.02) | 55.34 (13.32) | 49.30 (15.57) | <.001 |
| Sex, n (%) | | | | | <.001 |
| Male | 879 (84.4) | 377 (91.5) | 398 (82.1) | 104 (72.2) | |
| Female | 162 (15.6) | 35 (8.5) | 87 (17.9) | 40 (27.8) | |
| Rheumatic disease, n (%) | 4 (0.4) | 4 (1.0) | 13 (2.7) | 10 (6.9) | <.001 |
| EuroSCORE II (IQR) | 1.89 (1.77) | 1.64 (1.45) | 1.83 (1.11) | 1.83 (1.78) | <.001 |
| COPD, n (%) | 29 (2.8) | 13 (3.2) | 21 (4.3) | 13 (9.0) | .291 |
| Aortic valve morphology, n (%) | | | | | <.001 |
| Bicuspid | 459 (44.1) | 85 (20.6) | 320 (66.0) | 60 (41.7) | |
| Tricuspid | 578 (55.5) | 326 (79.1) | 163 (33.6) | 80 (55.6) | |
| Quadricuspid | 4 (0.4) | 1 (0.2) | 2 (0.4) | 4 (2.8) | |
| Type of surgery, n (%) | | | | | <.001 |
| AVP | 1031 (99.0%) | 374 (90.8) | 362 (74.6) | 114 (79.2) | |
| AVR | 10 (1.0%) | 38 (9.2) | 123 (25.4) | 30 (20.8) | |
| Valve repair technique, n (%) | | | | | <.001 |
| Isolated valve repair | 281 (27.0) | 155 (37.6) | 91(18.8) | 53 (36.8) | |
| Valve repair + partial root replacement | 32 (3.1) | 9 (2.2) | 35 (7.2) | 2 (1.4) | |
| Valve repair + VSRR | 548 (52.6) | 189 (45.9) | 156 (32.2) | 36 (25.0) | |
| Valve repair + tubular aorta | 170 (16.3) | 21 (5.1) | 80 (16.5) | 23 (16.0) | |
| Cusp repair technique, n (%) | | | | | |
| Plicating stitches | 778 (74.7) | 269 (65.3) | 204 (42.1) | 84 (58.3) | <.001 |
| Cusp resection | 141 (13.5) | 24 (5.8) | 101 (20.8) | 34 (23.6) | <.001 |
| Running suture | 55 (5.3) | 48 (11.7) | 52 (10.7) | 3 (2.1) | <.001 |
| Decalcification | 11 (1.1) | 4 (1.0) | 259 (53.4) | 7 (4.9) | <.001 |
| Patch reconstruction | 36 (3.5) | 115 (27.9) | 42 (8.7) | 23 (20.0) | <.001 |
| Annuloplasty technique, n (%) | | | | | |
| External ring | 219 (21.0) | 193 (46.8) | 107 (22.1) | 27 (18.8) | <.001 |
| Internal ring | 3 (0.3) | 0 (0.0) | 4 (0.8) | 1 (0.7) | .088 |
| Suture annuloplasty | 286 (27.5) | 60 (14.6) | 7 (1.4) | 22 (15.3) | <.001 |
| Cabrol stitches | 148 (14.2) | 50 (12.1) | 115 (23.7) | 23 (16.0) | <.001 |
| STJ ring | 25 (2.4) | 35 (8.5) | 15 (3.1) | 12 (8.3) | <.001 |

Values are mean (SD), n (% of subgroup), or median (IQR). *SD*, Standard deviation; *EuroSCORE*, European System for Cardiac Operative Risk Evaluation; *IQR*, interquartile range; *COPD*, chronic obstructive pulmonary disease; *AVP*, aortic valvuloplasty; *AVR*, aortic valve replacement; *VSRR*, valve-sparing root replacement; *STJ*, sinotubular junction.

**TABLE 2. Adverse events of patients undergoing surgical AVP and AVR at 2-y follow-up**

| | Postoperative complications | | | | |
|---|---|---|---|---|---|
| | AV-related, n (%) | Thromboembolic events, n (%) | Major bleeding, n (%) | Pacemaker implantation, n (%) | Other cardiac reoperations, n (%) |
| AVP (N = 1881) | | | | | |
| Control | 17 (1.6) | 10 (1.0) | 4 (0.4) | 8 (0.8) | 7 (0.7) |
| Fenestration | 9 (2.4) | 2 (0.5) | 4 (1.1) | 4 (1.1) | 5 (1.3) |
| Calcification | 11 (3.0) | 6 (1.7) | 3 (0.8) | 1 (0.3) | 6 (1.7) |
| Retraction | 8 (7.0) | 0 (0.0) | 1 (0.9) | 1 (0.9) | 0 (0.0) |
| AVR (N = 201) | | | | | |
| Control | 0 (0.0) | 1 (10) | 0 (0.0) | 0 (0.0) | 2 (20) |
| Fenestration | 0 (0.0) | 2 (5.3) | 0 (0.0) | 2 (5.3) | 0 (0.0) |
| Calcification | 3 (0.8) | 0 (1.0) | 1 (0.8) | 1 (0.8) | 1 (0.8) |
| Retraction | 1 (3.3) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) |

Values are n (% of subgroup cusp characteristics undergoing AVP/AVR). *AV*, Aortic valve; *AVP*, aortic valvuloplasty; *AVR*, aortic valve replacement.

**TABLE 3. Logistic generalized estimating equation analysis for switch to surgical aortic valve replacement for each specific cusp characteristic**

| | Switch to aortic valve replacement | |
|---|---|---|
| | OR (95% CI) | *P* value |
| Fenestration | 12.32 (4.47-33.90) | <.001 |
| Calcification | 13.50 (3.30-55.15) | <.001 |
| Retraction | 25.14 (4.84-130.57) | <.001 |

Adjustments for sex, age, EuroSCORE II and valve morphology at baseline have been made. *OR*, Odds ratio; *CI*, confidence interval.

crossclamping session. In the remaining 9 cases, surgeons switched to AVR after a second (5 cases) or third (4 cases) crossclamping session.

## Postoperative AR

Postoperative changes in AV function after AVP were evaluated using repeated measurements of AR degree. For every patient with AVP, both an echocardiogram at discharge and follow-up was provided, resulting in a total of 3762 echocardiograms for 1881 patients. Median follow-up time was 389 days (interquartile range, 488 days) following a right-skewed distribution. Table E1 provides the baseline AR grade for each cusp characteristic. Odds for developing moderate (grade 3) or severe AR (grade 4) compared with absent or mild AR (grade 0 or 1) postoperatively was investigated separately for each cusp characteristic (Table 4).

Findings demonstrate a significant trend toward a greater risk of developing severe AR (grade 4) after AVP for patients with cusp calcification and cusp retraction (OR, 6.69; 2.67-16.80; $P \leq .001$ and OR, 4.24; 1.12-16.11; $P = .034$) compared with the control group. These patients were more likely to develop grade 4 AR than grade 0 or 1 AR during a median follow-up period of 1.1 years. In addition, results show that both cusp retraction and fenestration are associated with increased likelihood of postoperative grade 2 AR compared with the control group (OR, 1.95; 1.25-3.02; $P = .003$ and OR, 1.41; 1.04-1.92; $P = .028$). Only patients having AV cusp retraction were associated with postoperatively developing grade 3 AR (OR, 2.81; 1.01-7.82; $P = .048$).

## AV-Related Reintervention

AV-related reintervention was performed in 45 of 1881 patients who underwent AVP (2.4%); 17 patients in the control group, 11 patients with cusp calcification, 9 with fenestration, and 8 with retraction. A significant log-rank Mantel–Cox test ($P = .003$) revealed substantial differences regarding AV-related reintervention between groups. Figure 3 shows significantly increased hazards of reintervention in the study group compared with the control group. Cox-regression analyses indicated that patients with cusp retraction had a significantly increased hazard ratio (HR) of requiring reintervention compared to the control group, both at 1 and 2 years follow-up (HR, 5.66; 2.19-14.67; $P \leq .001$ and HR, 3.09; 1.37-7.58; $P = .007$) (Table 5). Neither sex ($P = .818$), age ($P = .810$) or valve morphology ($P = .761$) were effect-modifiers for AV-related reintervention.

There was surgical variety in AVP procedures among the patients requiring AV-related reintervention (N = 45). In total, 24 patients underwent isolated valve repair, 15 patients were treated using a valve-sparing root replacement alongside valve repair, and the remaining 6 patients underwent valve repair combined with replacement of the tubular aorta (Table E2).

$\chi^2$ testing revealed no differences between cusp characteristics with regards to sex ($P = .234$), type of valve repair ($P = .943$), and valve morphology (bicuspid or tricuspid AV; $P = .077$) in this subgroup of patients. Kruskal–Wallis testing did not show any significant differences in EuroSCORE II, either ($P = .968$).

Interestingly, mean age was substantially lower in the group of reinterventions ($\mu = 43.89$; SD = 13.99) compared with the group with successful AVP procedures ($\mu = 53.53$; SD = 13.56). Binary logistic regression analysis revealed that increased age was associated with significantly decreased odds for AV-related reintervention at 2 years follow-up (OR, 0.95; 0.93-0.97; $P \leq .001$).

In addition, a "freedom from severe AR (grade 4)" analysis was performed using the Kaplan–Meier and Cox-regression method to determine AVP durability (Figure E2 and Table E3). A significantly increased hazard of severe AR at both 1 and 2 years' follow-up was reported for patients with cusp retraction (HR, 6.45; 1.73-24.01;

**TABLE 4. Pairwise generalized estimating equation analysis for aortic regurgitation grade over time for specific cusp characteristics**

| | Postoperative grade of aortic regurgitation for each specific cusp characteristic | | | |
|---|---|---|---|---|
| | Grade 3 | | Grade 4 | |
| | OR (95% CI) | *P* value | OR (95% CI) | *P* value |
| Fenestration | 1.46 (0.68-3.11) | .330 | 1.12 (0.28-4.49) | .875 |
| Calcification | 1.50 (0.71-3.18) | .290 | 6.69 (2.67-16.80) | <.001 |
| Retraction | 2.81 (1.01-7.82) | .048 | 4.24 (1.12-16.11) | .034 |

Adjustments for sex, age, EuroSCORE II, and time have been made. *OR*, Odds ratio; *CI*, confidence interval.



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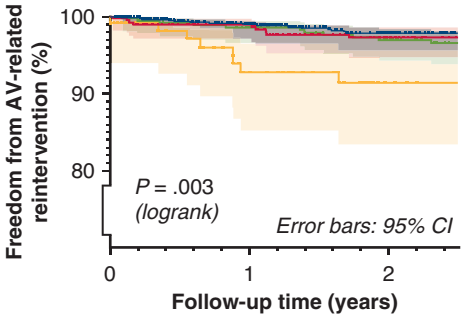


Patients at risk

| | Baseline | 1 year | 2 years |
|---|---|---|---|
| Control | 1031 | 777 | 652 |
| Fenestration | 374 | 309 | 251 |
| Calcification | 360 | 321 | 276 |
| Retraction | 114 | 81 | 58 |

**FIGURE 3.** Freedom from reintervention after aortic valvuloplasty for specific cusp characteristics. *AV*, Aortic valve.

$P = .005$ and HR, 5.00; 1.40-17.62; $P = .013$). Cusp calcification was associated with an increased hazard of severe AR at 2 years' FU (HR, 3.17; 1.12-9.00; $P = .030$).

## DISCUSSION

This study investigated whether specific cusp characteristics (retraction, calcification, and fenestration) affect clinical outcomes of AVP. As primary outcome variables, we explored perioperative switch to AVR, postoperative grade of AR, and time to AV-related reintervention between control and study groups.

Our study findings indicate that each of the studied AV cusp characteristics is associated with an increased likelihood of switching to AVR perioperatively despite a preoperative intention for AVP, compared with the control group. These results suggest that perioperative assessment plays an important role in determining the appropriate course of treatment for patients with these specific cusp characteristics. The majority of conversions to AVR (N = 59) were based on initial perioperative AV assessment. In 12 patients, a switch to AVR was performed after the second (N = 7) or third (N = 5) crossclamping session, primarily due to residual AR (N = 11). One patient developed aortic valve stenosis following AVP and was treated with AVR. Of these 12 patients, 4 had cusp retraction, 4 had cusp fenestration, and 2 cases each were attributed to the AV cusp calcification group and the control group.

Furthermore, our results indicate that cusp calcification and retraction are predictive factors for the development of severe AR after AVP. However, we did not find significant differences in the development of grade 3 AR over time in patients with cusp calcification in our GEE analysis, which may be attributed to the availability of only 2 echocardiograms per patient. One echocardiogram was obtained at discharge and the other was selected based on a predefined criterion: the highest grade of AR during follow-up. Therefore, any patient with severe AR would have met the criteria for moderate AR (grade 3) at some point during follow-up. Therefore, the observed differences between cusp characteristics may also be relevant for the development of grade 3 AR.

Overall prevalence of AV-related reintervention was low. Still, our results suggest that AVP should be performed with caution in patients with cusp retraction. Of 114 patients with cusp retraction, 8 patients required reintervention within 2 years' follow-up. Five of these patients (62.5%) were treated using a patch repair (cusp extension = 3; cusp belly = 2) at the initial AVP. The total number of patch repairs in the cusp retraction group was 23, implying that 21.7% of these interventions was unsuccessful. Several authors previously proposed that patch repair is associated with AR recurrence in patients with AV cusp retraction.[16,27,28] Our results support the notion that cusp retraction could be a (relative) contraindication for AVP using patch repair.

Interestingly, when adding up the aforementioned results, it appears that surgeons are well able to distinguish between AVP feasibility in patients with AV cusp fenestration. Even though all cusp characteristics showed increased odds of switching to AVR, patients with AV cusp fenestration were the only group showing neither an increased risk of severe AR recurrence ($P = .57$) or early reintervention ($P = .88$) compared with the control group. In patients with AV cusp retraction or calcification, it seems that patients too often undergo AVP resulting in early related reintervention or recurrence of severe AR.

**TABLE 5. Hazard ratios for reintervention at 1 and 2 years' follow-up for specific cusp characteristics**

| | Aortic valve-related reintervention following aortic valvuloplasty | | | |
|---|---|---|---|---|
| | 1-y follow-up | | 2-y follow-up | |
| | HR (95% CI) | *P* value | HR (95% CI) | *P* value |
| Fenestration | 1.31 (0.43-3.94) | .636 | 1.28 (0.55-2.94) | .566 |
| Calcification | 1.25 (0.42-3.72) | .689 | 1.67 (0.77-3.63) | .193 |
| Retraction | 5.66 (2.19-14.67) | <.001 | 3.22 (1.37-7.58) | .007 |

Adjustments for sex, age, valve morphology, and EuroSCORE II have been made. *HR*, Hazard ratio; *CI*, confidence interval.

Future research should focus on investigating which other variables could be of influence in this association. Rheumatic diseases, for instance, are linked to disease progression in patients with cusp retraction and calcification.[29,30] This could therefore partly explain why patients with these specific characteristics have greater odds of developing severe AR during follow-up. It would be insightful to stratify outcomes for different cusp characteristics according to the specific used AVP procedure. Differences in objective parameters such as geometric height should be included in these studies as well, as this could create appropriate cut-off points for AVP feasibility. In this study, geometric height data was not reported consistently and frequently enough to consider.

A substantial number of patients was excluded due to limited echocardiographic data availability, possibly inducing selection bias. This is especially noticeable when comparing the subgroup of cusp retraction with the total sample size (cusp retraction = 144; 6.9% of total sample). Preferably, study groups would have been more comparable in size. Due to these differences, alongside the relatively low number of events in this population, the confidence intervals of our analyses are wide as well. However, considering used matching procedures, adjustments for possible confounding and similarities with findings reported by previous authors,[12-16] it is justified to assume that possible interference in this study was minimal.

One of the issues observed in this study is the variation in follow-up time for echocardiograms between different centers. This variance can be attributed to the absence of a standardized protocol for all centers. To overcome this problem in the future, it is crucial to establish a standard protocol that all centers follow to ensure consistent and comparable results across different groups.

Another issue that could have contributed to variance in follow-up time is the inadequate updating of patient data into the database, resulting in incomplete patient files. This led to the exclusion of some patients from the analysis, which can have a significant impact on the study's overall findings. Frequent and accurate data input is critical to ensure that patient data are up-to-date and readily available for analysis, enabling researchers to obtain more accurate and reliable results. By adopting these practices, we can ensure that future studies have consistent and reliable data, enabling us to make informed decisions about patient care.

As can be deduced from "Switch to AVR" analysis in the Results section, the ORs and corresponding 95% confidence intervals do not appear to be biologically credible. This could partially be explained by collinearity and the limited number of events. Perioperative conversion to AVR was performed only in 3.6% of cases spread across 4 groups. However, these results do underline the notion that abnormal cusp characteristics could

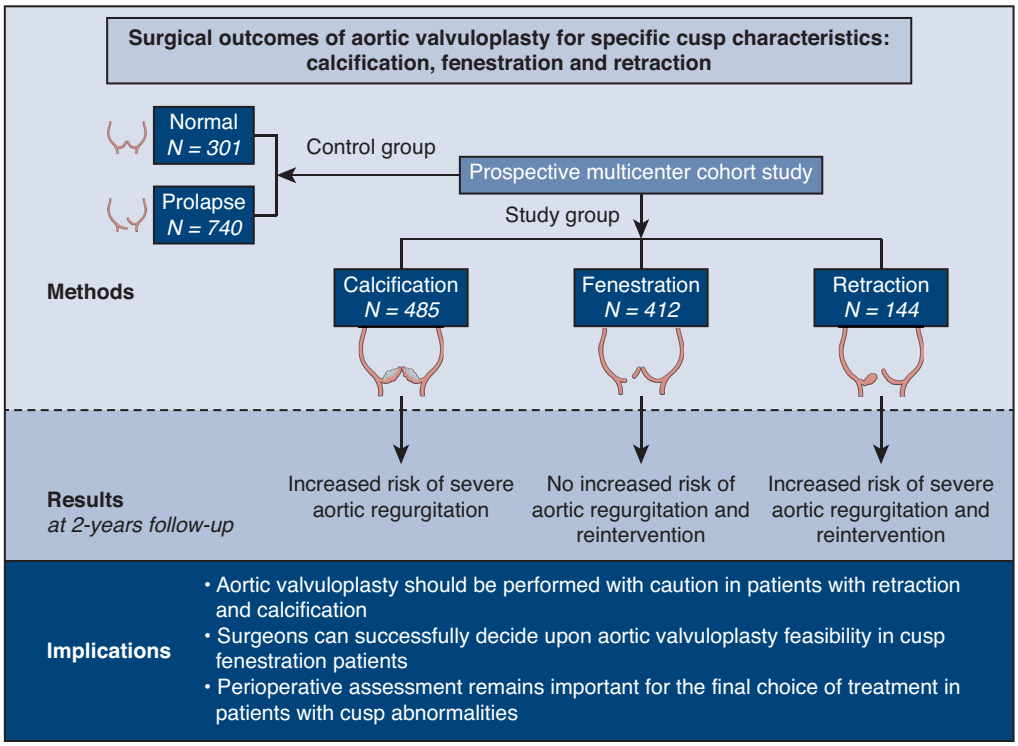


**FIGURE 4.** Surgical outcomes of aortic valvuloplasty for specific aortic valve cusp characteristics; retraction, calcification and fenestration.

affect the feasibility or rather, the choice to perform an AVP.

Another limitation involves data-clustering. The results in this paper are based on data from a multicenter database, accounting for possible differences between hospitals with regards to patient population and care. We minimalized this interference with a subject-variable at center-level. However, we were unable to do so for the time to AV-related reintervention and postoperative AR analyses due to nonconvergence of the statistical models.

Lastly, it should be noted that the study period is relatively long; 26 years. It is presumable that developments in perioperative care, indications for surgery, and surgical techniques over the years could have gradually affected the outcomes.

## CONCLUSIONS

Patients with cusp retraction appear to be less suitable candidates for AVP in the short term. These patients show an increased risk for developing postoperative AR and increased hazards of AV-related reintervention at 1 and 2 years' follow-up. Patients with cusp calcification show increased odds of developing postoperative severe AR (Figure 4).

Contrary, patients with cusp fenestration showed excellent feasibility for AVP, without increased odds for AV-related reinterventions or postoperative severe AR, despite an increased likelihood of perioperative switching to AVR. These results suggest that surgeons can appropriately distinguish between feasibility for AVP and AVR when encountering AV cusp fenestration. Future studies should focus on providing more insight into the influence of different AV cusp characteristics on AVP feasibility, as this could potentially contribute to the development and implementation of new guidelines for AR treatment.

### Conflict of Interest Statement

The authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

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**Key Words:** aortic valve repair, aortic regurgitation, valve cusp characteristics, calcification, fenestration, retraction

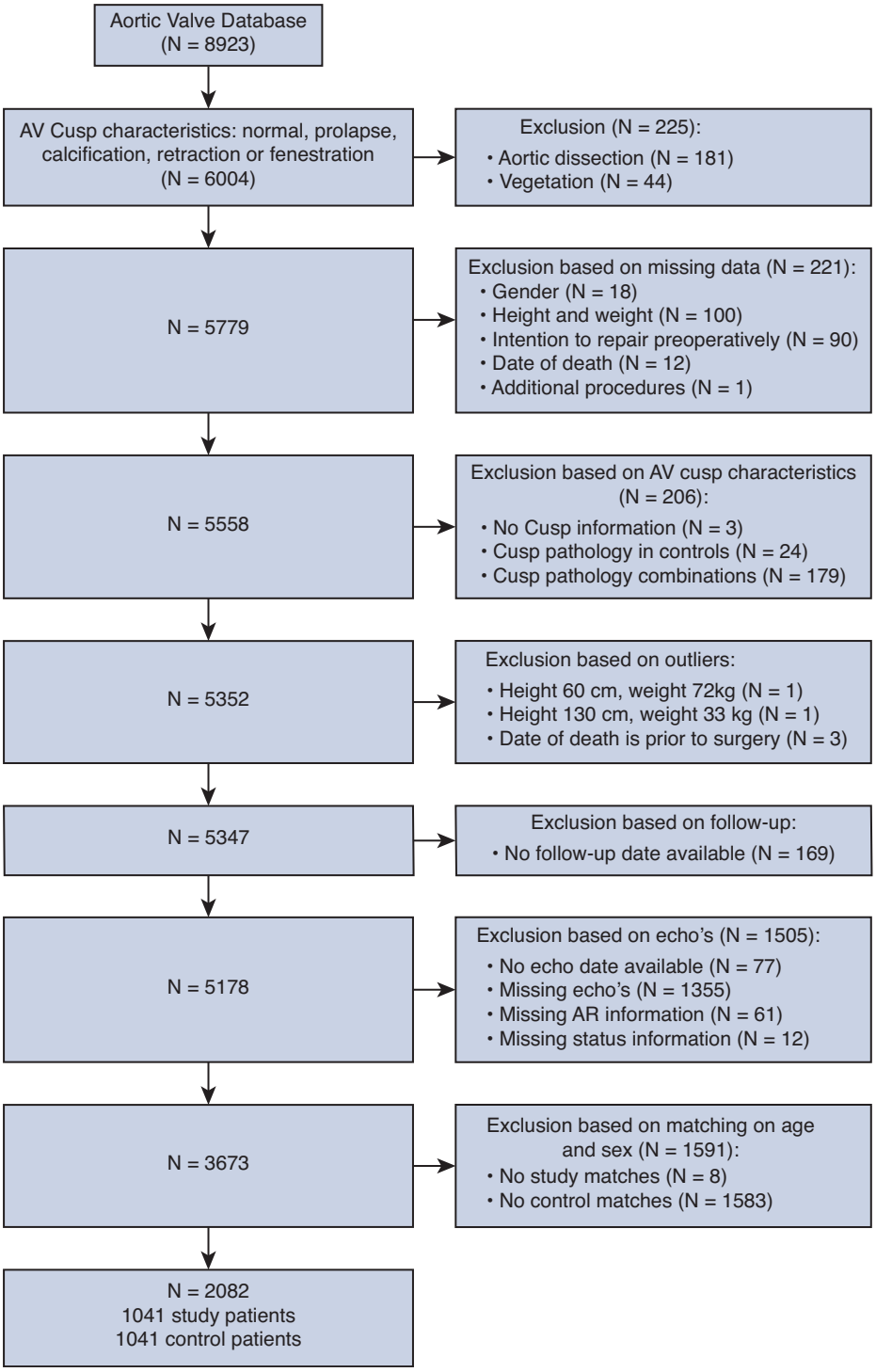
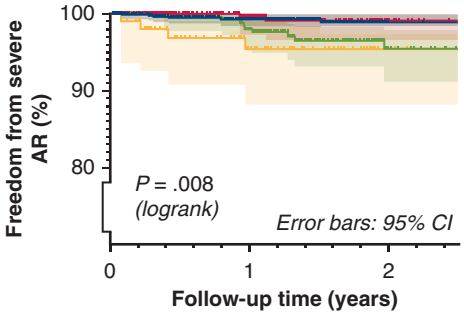



**FIGURE E1.** Flowchart showing selection process of study participants. *AV*, Aortic valve.




**Patients at risk**

| | *Baseline* | *1 year* | *2 years* |
|---|---|---|---|
| Control | 1031 | 493 | 26 |
| Fenestration | 374 | 230 | 75 |
| Calcification | 362 | 232 | 83 |
| Retraction | 114 | 64 | 25 |

**FIGURE E2.** Freedom from severe postoperative AR after aortic valvuloplasty for specific cusp characteristics. *AR*, Aortic regurgitation; *CI*, confidence interval.

**TABLE E1. Grade of AR at baseline for specific cusp characteristics**

| AR at discharge | Control (N = 1031) | Fenestration (N = 374) | Calcification (N = 362) | Retraction (N = 114) |
|---|---|---|---|---|
| 0—absent (%) | 567 (55.0) | 171 (45.7) | 219 (60.5) | 53 (46.5) |
| 1—mild (%) | 417 (40.4) | 181 (48.4) | 121 (33.4) | 48 (42.1) |
| 2—mild to moderate (%) | 46 (4.5) | 21 (5.6) | 22 (6.1) | 11 (9.6) |
| 3—moderate to severe (%) | 1 (0.1) | 1 (0.3) | 0 (0.0) | 2 (1.8) |
| 4—severe (%) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 0 (0.0) |

*AR*, Aortic regurgitation.

**TABLE E2. Operative characteristics of patients requiring reintervention following AVP**

| | Surgical characteristics for patients requiring reintervention after AVP | | | |
|---|---|---|---|---|
| | Control (N = 17) | Fenestration (N = 9) | Calcification (N = 11) | Retraction (N = 8) |
| Valve repair technique | | | | |
| Isolated valve repair | 8 (47.1%) | 5 (55.6%) | 6 (54.5%) | 5 (62.5%) |
| Valve repair + VSRR | 8 (47.1%) | 4 (44.4%) | 2 (18.2%) | 1 (12.5%) |
| Valve repair + tubular aorta | 1 (5.9%) | 0 (0.0%) | 3 (27.3%) | 2 (25.0%) |
| Cusp repair technique | | | | |
| Plicating stitches | 14 (82.4%) | 7 (77.8%) | 6 (54.5%) | 1 (12.5%) |
| Cusp resection | 1 (5.9%) | 0 (0.0%) | 4 (36.4%) | 3 (37.5%) |
| Running suture | 0 (0.0%) | 0 (0.0%) | 4 (36.4%) | 0 (0.0%) |
| Decalcification | 1 (5.9%) | 0 (0.0%) | 9 (81.8%) | 1 (12.5%) |
| Patch reconstruction | 3 (17.6%) | 2 (22.2%) | 5 (45.5%) | 5 (62.5%) |
| Annuloplasty technique | | | | |
| External ring | 5 (29.4%) | 4 (44.4%) | 0 (0.0%) | 0 (0.0%) |
| Suture annuloplasty | 6 (35.3%) | 2 (22.2%) | 1 (9.1%) | 2 (25.0%) |
| Cabrol stitches | 3 (17.6%) | 1 (11.1%) | 6 (54.5%) | 1 (12.5%) |

Values are n (%) of subgroup cusp characteristics undergoing AVP. *AVP*, Aortic valvuloplasty, *VSRR*, valve-sparing root replacement.

**TABLE E3. HRs for severe aortic regurgitation at 1 and 2 years' follow-up for specific cusp characteristics**

| | Severe aortic regurgitation (grade 4) following aortic valvuloplasty | | | |
|---|---|---|---|---|
| | 1-y follow-up | | 2-y follow-up | |
| | HR (95% CI) | *P* value | HR (95% CI) | *P* value |
| Fenestration | 0.46 (0.05-4.0) | .483 | 0.71 (0.14-3.55) | .680 |
| Calcification | 2.32 (0.67-8.02) | .183 | 3.17 (1.12-9.00) | .030 |
| Retraction | 6.45 (1.73-24.01) | .005 | 5.00 (1.40-17.62) | .013 |

*HR*, Hazard ratio; *CI*, confidence interval.